

SEC P vs NP log 2026-05-12

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 01:30 UTC

SEC P vs NP — daily log 2026-05-12

1 cycles recorded

Free Cumulant Sum Bounded by Disjointness Communication Complexity

- **Verdict:** INCONCLUSIVE
- **Bridge:** Free Probability Theory $\times$ Communication Complexity of DISJOINTNESS
- **Recorded:** 2026-05-12 01:18 UTC
- **Entry ID:** ecd8f062c47d

Statement

For a $|X| \times |Y|$ matrix M defining a partial function f , let $\mu(M) = \sum_{k \geq 1} |c_k(M)|$ where $c_k(M)$ are the free cumulants of M 's entries. Then $\mu(M) \leq O(\sqrt{R(f)})$ where $R(f)$ is the randomized communication complexity of f . For DISJ_n , $\mu(\text{DISJ}_n) = \Omega(n)$ holds.

Rationale

Free cumulants capture non-commutative independence structures in matrix entries, which may correlate with the information-theoretic cost of coordinating disjointness checks. The sum's growth rate could reflect the minimal communication required to resolve entangled dependencies.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.09s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 75, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 54, in run_trial
    cumulants = free_cumulants(M)
                ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_38ecf086.py", line 39, in free_cumulant
    R[i][j] -= sum(R[x][y] * R[i - x][j - y] for x in range(i) for y in range(j)) / (i * j)
               ^~~~~~
```

ZeroDivisionError: division by zero

Judge reasoning

Test crashed due to division by zero, preventing evaluation of conjecture | next: Fix implementation to handle zero division and retest with validated inputs